

PQC in DNSSEC: Securing the Internet's Phonebook

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DAILY MONITORING

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While TLS, SSH, and VPNs get much of the PQC attention, securing the Domain Name System (DNS) is equally critical. DNS Security Extensions (DNSSEC) provide authenticity and integrity for DNS responses, preventing attacks like cache poisoning and redirection. However, the classical algorithms used in DNSSEC (RSA, ECDSA) are vulnerable to quantum attacks. This study explores the transition to PQC in DNSSEC.

The Challenge: Signature Size and Network Overhead

The primary challenge with integrating PQC into DNSSEC is the size of the digital signatures. DNS responses are typically sent over UDP, which has strict packet size limitations. Large PQC signatures can lead to fragmentation or force a fallback to TCP, both of which introduce latency and performance degradation. This makes stateless hash-based signature schemes like SLH-DSA (SPHINCS+) and lattice-based schemes like ML-DSA (CRYSTALS-Dilithium) the main candidates, each with distinct trade-offs.

Why is this crucial for Internet Security?

An insecure DNS means an attacker can redirect users from legitimate sites (e.g., your bank) to malicious phishing sites, even if the target site uses perfect TLS. Securing DNS with quantum-resistant algorithms ensures that the very first step of accessing a website—looking up its address—is trustworthy in the quantum era.

Comparing PQC Candidates for DNSSEC

The IETF is actively debating which PQC algorithms are best suited for DNSSEC. The choice involves a careful balance between security, signature size, and verification speed.

Algorithm	Type	Key/Signature Size	Pros	Cons
ML-DSA (Dilithium)	Lattice-Based	Moderate	- Fast signing & verification - Relatively small signatures	- Newer cryptographic assumptions
SLH-DSA (SPHINCS+)	Hash-Based	Large Signatures, Small Keys	- Very strong security assumptions (depends only on hash function security) - Small public keys	- Very large signatures (~8 KB to ~49 KB) - Slower signing process - Stateful signing limitations

For DNSSEC, ML-DSA is emerging as a strong contender due to its smaller signature sizes compared to SLH-DSA, making it more practical for UDP-based protocols. The IETF draft [draft-ietf-dnsop-dnssec-pcurves](#) is already paving the way for new algorithms, and similar standardization for PQC is in progress.

Daily Quiz

- **1.** What is the primary technical challenge when implementing PQC algorithms in DNSSEC?
- **2.** Why is SLH-DSA (SPHINCS+) considered highly secure but problematic for DNSSEC?
- **3.** Which NIST-standardized PQC algorithm is currently seen as a more practical fit for DNSSEC, and why?